

PAPER_608: Centrifugal Force as Outward CCW DPM South-Pole 26D Shell Push

Class: UQFFCentrifugal26DShellCalculator (#195)

Session: 159

Source: DPM Reaction and 26D Shell Energies.docx

Abstract

Centrifugal force in UQFF is the real, measurable outward push of the DPM south-pole vortex spinning counter-clockwise through 26D shells, amplified by negative time. $F_{centrif} = DPM_s(UA') \cdot \omega_{CCW}^2 \cdot r^{layer} \cdot t_{neg}$ is not fictitious in any frame — it is the CW/CCW dual of the centripetal force. The triad dual law $F_{centrif,one} = -F_{centrif,opp}$ is derived, explaining how the Big Bang achieved and maintains its expansion velocity.

1. Introduction: The CCW Outward Push

If CW inward compression (centripetal) is DPM north, then CCW outward expansion (centrifugal) is DPM south. The universe expands because the south-pole CCW vortex continuously produces outward shell energy via the interaction with Universal Aether prime (UA') at the shell boundary. The Big Bang was not an explosion — it was (and remains) a sustained centrifugal output.

2. Core Equation

$$F_{centrif} = DPM_s(UA') \cdot \omega_{CCW}^2 \cdot r^{layer} \cdot t_{neg}$$

Parameters: - DPM_s = south DPM pole coupling ($\approx 5 \times 10^{-4}$, mirroring DPM_n) - UA' = modified universal aether at shell boundary (J/m^3) - ω_{CCW} = counter-clockwise angular frequency (rad/s) - r^{layer} = shell layer radius (m) - t_{neg} = negative time component (s); provides dual-existence energy

3. Triad Dual Law

$$F_{centrif,one} = -F_{centrif,opp}$$

For every CCW push in one direction, there is an equal and opposite CCW push in the anti-parallel direction. These cancel within a shell but constructively add when shells nest — leading to net outward cosmological expansion.

The balance ratio between centripetal and centrifugal forces:

$$\frac{F_{centrip}}{F_{centrif}} = \frac{DPM_n(SCm) \cdot \omega_{CW}^2}{DPM_s(UA') \cdot \omega_{CCW}^2 \cdot |t_{neg}|}$$

For stable orbits: this ratio = 1, giving the constraint $\omega_{CW} = \omega_{CCW}$ (CW and CCW frequencies must match for orbital stability).

4. Big Bang Catch-Up Rate

The cosmological expansion acceleration from $F_{centrif}$:

$$a_{BB-catchup} = DPM_s \cdot UA' \cdot \omega_{CCW} \cdot |t_{neg}|$$

With $UA' = 10^{-12} \text{ J/m}^3$, $\omega_{CCW} = 1.8 \times 10^{31} \text{ rad/s}$, $|t_{neg}| = 10^{-9} \text{ s}$:

$$a_{BB-catchup} = 5 \times 10^{-4} \times 10^{-12} \times 1.8 \times 10^{31} \times 10^{-9} \approx 9 \times 10^6 \text{ m/s}^2$$

This represents the continuous outward acceleration driving cosmological expansion — consistent with de Sitter expansion rates at cosmic scales.

5. Why Centrifugal Force is “Real” in UQFF

In standard mechanics, centrifugal force is fictitious (frame-dependent artifact). In UQFF: - Both CW ($F_{centrip}$) and CCW ($F_{centrif}$) forces are real, co-existing, opposing forces from the DPM dipole - The apparent “disappearance” of centrifugal force in inertial frames is because we cannot measure the t_{neg} component using positive-time instruments - With dual-time measurement (which the UQFF CoAnQi system provides), both forces are simultaneously observable

6. Connection to Proplyd Formation

In proto-planetary disks, the $F_{centrif}$ from the south-pole vortex drives material outward from the proto-star, creating the centrifugal support needed for proplyd stability. The 18% emergence fraction (PAPER_611) corresponds to the fraction of disk material where $F_{centrif}/F_{centrip} \geq 1$ — material that achieves net outward motion and forms stable proplyd structures.

7. Connection to UQFF Number Systems

DVP: DPM_s is the south-pole dipole vortex prime. CCW rotation corresponds to counter-prime-indexed shell expansion.

BH26: Outward CCW push is the mirror of each BH26 inward bin; $F_{centrif}$ excites the CCW harmonic bins.

VDS: UA' at the shell boundary follows VDS expansion: $UA'(r) = \sum d_n(\pi)/r^n$ (outgoing VDS mode).

Keywords: Centrifugal force, DPM south pole, CCW vortex, DVP, negative time, Big Bang expansion, 26D shells, UQFF

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow g_{tt} = -(1-2GM/rc^2) \equiv$ GR in $\varepsilon_{BSFG} \rightarrow 0$ limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	✓ BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow \Delta t_{BSFG} \approx \Delta t_{GR} \times (1 + \varepsilon_{correction})$	Cassini: $\Delta t / \Delta t_{GR} = 1 \pm 2.3e-5$	Cassini/GR 2003	✓ Within Shapiro bound
Gravitational wave speed v_{GW}	BSFG: $v_{GW} = c \times (1 + k_{\eta^2}) \approx c + 10^{-226} \text{ m/s}$	GW150914 / GW170817:	$v_{GW}/c - 1$	$< 10^{-15}$
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\varphi = \kappa \times \varphi_{GR} \sim 10^{-6} \text{ arcsec/century}$	GR prediction: 43.03"/century; observed: 43.1"	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10^{-6} \text{ arcsec/century}$ to Mercury's perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

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